Final Report of the Project ar4wind

Supporting Citizen Participation Processes Through Ad-Hoc Visualization of Planned Wind Energy Turbines Using Mobile Augmented Reality Technology (mAR)

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1. Part I: Brief Project Overviews

1.1. Task Description

Onshore wind energy represents a fundamental pillar for achieving national climate protection and greenhouse gas mitigation goals in Germany. However, deciding on specific spatial locations for future wind farm installations frequently triggers protracted and highly controversial debates at the local municipal level. Critics primarily focus on potential acoustic emissions, landscape transformation, and visual impacts on regional scenery. To ensure highly transparent communication with affected local citizenries and foster enhanced project acceptance, practical and interactive visualization tools can be successfully integrated directly into municipal public participation frameworks.

The overarching core objective of the ar4wind applied research project is to comprehensively investigate the exact extent to which qualified, fact-based spatial debates regarding wind energy proposals can be positively supported by localized visualizations of wind turbines displayed on consumer mobile devices like smartphones and tablets. The pivot of this research is the systematic deployment of mobile Augmented Reality (mAR) technology. Within informal citizen participation environments, planned wind energy facilities are rendered fully experiential as interactive 3D objects, embedded location- and reality-accurately directly into the real-time live camera feed of a mobile device. Throughout this project, the developed mAR system was directly evaluated under authentic field conditions regarding its operational workflow suitability within everyday planning routines and its communicative effectiveness inside formal and informal public dialogue processes. Early informal phase visualization instruments provide a constructive foundation for multi-stakeholder dialogue, allowing diverse interest groups to align and communicate transparently.

1.2. Scientific and Technical State of the Art

Visualizations in Public Participation for Energy Infrastructure Projects

The interactive and productive integration of citizens into the spatial execution of the national energy transition remains a profound socio-political challenge. Traditional planning and approval procedures regularly carry extreme social conflict potential, which is heavily amplified by insufficient, opaque, or overly technical early public communication. Integrating interactive visualizations that conform to rigorous spatial quality criteria within structured community dialogues can effectively anchor highly polarized debates onto objective facts. These tools enable straightforward evaluations of potential spatial options and effortlessly demonstrate complex visual impacts, such as sightline obstructions, shifts in the local landscape character, and periodic shadow casting effects. Socio-ecological research indicates that sophisticated participation frameworks leveraging highly accurate visualization technologies significantly alleviate structural conflict potentials. This holds particularly true when interactive visual elements are deployed dynamically within flexible, informal municipal participation procedures early in the timeline.

Existing Wind Turbine Visualization Frameworks

In standard commercial planning practice, highly accurate photorealistic visualizations of large infrastructure installations are still predominantly generated statically on desktop workstations by specialized technical

layout experts. While these specialized rendering pipelines deliver highly realistic visual results of planned construction scenarios, they are inherently rigid. They must be prepared far in advance by technicians and are strictly constrained to pre-calculated view perspectives, specific field-of-view parameters, or fixed observation points (static photomontages). Alternatively, they require an entirely immersive virtual reality (VR) simulation cube, which disconnects the participant from the real ambient environment.

Geobased mAR Implementations and Spatial Geo-Localization

Developing applications that achieve realistic, stable AR overlays of virtual objects positioned kilometres away at explicit global coordinates is a highly complex engineering problem and a focus of active computer vision research. This difficulty is primarily caused by the low spatial accuracy and structural instability of standard built-in consumer hardware sensors (such as GPS chips and digital compasses). Pixel-accurate spatial AR tracking over large distances requires that the device position and 3D orientation (global heading vectors) are precisely calibrated against geographic coordinate systems immediately prior to use (Geo-Calibration). Furthermore, an outdoor mAR engine must simulate multi-layered occlusions in real time, ensuring that distant virtual items are correctly obscured by real foreground items like dense forest layers, rolling topography, or existing building facades. While specialized image recognition pipelines matching local scenery to expansive pre-registered 3D point clouds exist to counter these issues, they remain exceptionally compute-intensive, prone to local point matching failures, and difficult to scale across broad geographies.

mAR Visualization Tools for Wind Energy Infrastructures

At the time of the initial grant application, no commercial mAR software solution existed on the market capable of rendering distant wind turbines in an immersive, reality-compliant manner. Within academic literature and research repositories, only three early prototyping frameworks were documented: the "MoDal-MR" project by LandPlan OS GmbH, the "mARGo" software suite developed at HTW Berlin, and the "Linthwind" prototype created via a joint Swiss venture between Echtzeit GmbH and ZHAW. However, all three systems were limited to primitive prototype demonstrations containing basic functional tasks, and their actual suitability within public participation contexts had never been systematically assessed or validated through structured field studies.

1.3. Project Flow

The systematic execution of the ar4wind research timeline was structurally organized into six tightly integrated work packages (WPs) punctuated by five core milestones (M1–M5), as detailed below:

- **WP1 (Needs and Potential Analysis):** Conducted at the project's inception, this package focused on analyzing public participation methodologies across onshore wind planning cases, isolating structural weaknesses in current dialogue formats, and identifying specific entry vectors for mAR tech. FA Wind spearheaded the institutional process analysis, while HTW Berlin led the technical mAR potential matrix compilation.
- **WP2 (Representation Accuracy and Robustness Research):** Focused on evaluating existing mobile positioning tracking arrays and engineering a robust, geodata-driven outdoor GeoAR calibration framework.

- **WP3 (AR Visualization Methodologies):** Centered on creating high-performance visual rendering routines, exploring advanced occlusion algorithms, and integrating digital surface matrices.
- **WP4 (Technical Software Integration and Application Development):** Involved combining the algorithmic components into an intuitive, functional mobile app prototype optimized for end-user usability.
- **WP5 (Field Deployments and Optimization):** Consisted of executing two comprehensive field trial sequences and associated socio-scientific impact evaluations during live municipal planning scenarios.
- **WP6 (Knowledge Transfer, Coordination, and Project Management):** Led by EPC gGmbH, this continuous package ensured project governance, cross-partner tracking, public relations management, and target achievement.

Figure 1: Methodological Project Schema and Work Package Layout

The project structurally advanced from early WP1 stakeholder and technical requirements definition (Milestone 1) directly into parallel research tracks for positioning robustness (WP2) and visual rendering accuracy (WP3). These tracks converged inside WP4 for iterative prototype assembly across Milestone 2 and Milestone 3. Authentic community deployment loops occurred inside WP5 across Milestone 4, systematically backed by cross-cutting project management, public dissemination, and result transfer loops executed inside WP6 up to final Milestone 5 closure.

1.4. Key Results

The collaborative execution of the ar4wind research program successfully yielded the following primary technical and methodological outcomes:

1. **Algorithmic Optimization of Existing GeoAR Positioning Arrays:** The project successfully adapted and combined the foundational geodata rendering systems developed by HTW Berlin and LandPlan OS into a high-performance, unified mobile graphics pipeline. This included developing sequential, wizard-guided user interface workflows that simplify complex hardware sensor calibration tasks for non-technical users.
2. **Cross-Platform Technical Migration:** The underlying geodata-driven terrain positioning frameworks were successfully migrated onto a unified, commercial-grade real-time 3D development platform (the Unity engine), enabling stable deployment during the second field trial phase.
3. **Engineering of Distinct Structural Prototype Variants:** To test different technical approaches, the consortium engineered two functional mobile prototype applications utilizing distinct calibration methodologies:
    - *Application 1 (Prototype 2B):* Implements a terrain-matching GeoAR calibration method by dynamically loading and aligning 3D geodata and digital surface models.
    - *Application 2 (Prototype 2A):* Utilizes a manual object-referencing calibration model paired with an algorithmic, color-based sky segmentation mask for foreground occlusion rendering.

4. **Socio-Scientific Public Participation Field Insights:**
    - A remarkably simple, clean, and highly structured mobile user experience layout is paramount to achieving user adoption and positive engagement among citizens.
    - User onboarding must utilize step-by-step sequential calibration wizards, embed highly concise helper tooltips, and conceal professional expert configurations from normal users.
    - Operational deployment inside public participation environments strictly requires active moderation and expert technical guidance. The tool is not designed to be distributed to unassisted individuals lacking spatial planning knowledge.
    - Enabling authentic, real-time ad-hoc visual adjustments of turbine properties directly in the field substantially elevates the perceived credibility of and public trust in the digital planning data.

1.5. Cooperation with Other Entities

The execution of this research relied on continuous collaboration among the four core consortium partners: EPC gGmbH, FA Wind, HTW Berlin, and LandPlan OS GmbH. This core structure was systematically augmented by key practical field partners who provided logistically vital field validation opportunities. Notable field contributors included the municipal administration of the City of Bornheim (led directly by the Mayor and the Department of Civil Engineering), the specialized mediation consultancy *team ewen* (expert conflict and process managers), regional renewable project developers *Enertrag AG* and *Naturwind GmbH*, and the Regional Planning Association of West Mecklenburg.

2. Part II: Detailed Presentation

2.1. Overview of Achieved Results Against Targets

The principal objective of this research initiative was to engineer, optimize, and evaluate a field-ready mobile AR visualization platform for real-time, on-site simulations of projected wind farms. It also aimed to systematically analyze the socio-communicative opportunities, limits, and procedural requirements for embedding such tools within public participation frameworks. The completed project deliverables correspond directly with the predefined work plan targets across all core phases:

Analytical Phase Milestones

Predefined Project Target	Actual Achieved Project Outcome
1) Dialogue Needs Analysis: Isolate functional deficiencies and communicative challenges in current public dialogue tracks regarding the visual impact of wind turbines.	Concluded twelve comprehensive, problem-centered qualitative expert interviews across four core actor domains (Municipalities, Regional Planning, Project Developers, and Dialogue Mediators). The resulting insights, along with a detailed baseline paper mapping formal participation timelines, legal competencies, and content constraints across regional, land-use, and approval levels, were successfully delivered to the consortium.
2) Outdoor AR Tech Matrix: Classify and document the precise technical capabilities, physical limits, and sensor deficiencies of consumer-grade mobile outdoor AR.	Compiled an exhaustive mAR Technology Potential Matrix. This technical document detailed ambient tracking limitations, sensor drifts, and precise solution criteria for large-scale outdoor georeferencing and occlusion handling, which was distributed to all engineering tracks between September and November 2021.
3) Technical Requirements Co-Design: Run stakeholder workshops to define functional app features and user requirements.	Successfully organized and executed an interactive online Requirements Co-Design Workshop on November 24, 2021, with 20 key industry stakeholders. Technical concepts were disseminated via an introductory lecture, followed by structured whiteboarding sessions that produced a comprehensive functional requirements specifications document. This document was published on the project portal.

Technical Development Phase Milestones

Predefined Project Target	Actual Achieved Project Outcome
4) Spatial Calibration Framework: Research and deploy highly accurate, robust outdoor localization methods utilizing 3D geodata.	Evaluated existing internal tracking systems through rigorous multi-environment functional testing. Engineered a high-performance terrain-matching calibration routine driven by digital 3D surface models, validated during an extensive urban and rural field study with diverse test cohorts in July 2022. This framework was successfully migrated

Predefined Project Target	Actual Achieved Project Outcome
	onto a unified Unity production pipeline, rendering it fully operational for Phase II field deployments.
5) Realistic Occlusion Solutions: Devise advanced graphic systems to display planned wind energy infrastructure realistically, focusing on complex background/foreground occlusions.	Successfully integrated advanced occlusion rendering techniques into a unified Unity rendering script. Initial attempts to integrate acoustic emission audio profiles were abandoned due to disproportionate technical computing overheads. However, the team successfully engineered dual occlusion pathways: one driven by real-time 3D digital surface matrices, and another by real-time color-based sky segmentation filters. These methods allow vegetation and buildings to naturally obscure distant turbines, as validated in Phase II trials.
6) Iterative Prototype Engineering: Produce functional prototype applications for successive field testing phases, culminating in an optimized, highly usable system layout.	Delivered Prototype 1 in April 2022, built upon foundational code from LandPlan OS to anchor initial Phase I field trials. Based on user experience tracking, the team completely re-engineered the user interaction layer and produced two distinct, UI-optimized version 2 applications: Prototype 2A (integrating manual object referencing and sky segmentation) and Prototype 2B (integrating 3D geodata terrain matching). Due to extreme architectural differences in the underlying software backends and time constraints, merging both systems into a single composite application was not feasible within the project's lifespan.

Empirical Field Evaluation Phase Milestones

Predefined Project Target	Actual Achieved Project Outcome
7) Field Trial Executions: Deploy successive software prototypes across authentic planning regions to track operational utility.	Executed Practical Test Series I using Prototype 1 via four intensive field tours in Oederan (Saxony), Freigericht (Hesse), Bornheim (North Rhine-Westphalia), and Gehrden (Lower Saxony), gathering detailed survey metrics from approximately 100 participants. Executed Practical Test Series II across sites in Bruchsal, Freigericht/Alzenau, and a specialized field trial in the Berlin Wuhlheide forest.
8) Participatory Best Practices: Derive concrete recommendations for action to effectively embed AR tools within public participation processes.	Analyzed all field trial data and refined the insights during a professional Dialogue Workshop in January 2023 with 30 expert practitioners. The team demonstrated that rigorous pre-intervention stakeholder mapping and conflict analyses are vital prerequisites. They isolated key deployment parameters (such as group size limitations and structural differences between ad-hoc field AR versus pre-rendered field videos) and synthesized these insights into a formal guide for practitioners.
9) Engineering Optimization Loops: Evaluate technical field data to continuously improve application usability and UI design.	Identified extreme user friction during initial trials due to technical calibration complexities. Consequently, the team introduced a highly structured, sequential step-by-step calibration wizard, added concise overlay tutorials, and hid expert setup controls from regular users. This drastically enhanced user experience and layout responsiveness during subsequent independent internal testing loops.

Knowledge Dissemination and Administration Milestones

Predefined Project Target	Actual Achieved Project Outcome
10) Knowledge and Result Transfer: Disseminate scientific and practical findings to the professional public and maintain a project website.	Maintained consistent dissemination channels throughout the project. Technical advancements were published and presented at key national geographic symposia (such as the UIS Workshop 2022 and GeoForum MV 2022) and international computer science conferences (ICICT 2022; GISTAM 2022). Industry-specific findings were circulated through regular FA Wind newsletters, dedicated booths at major trade fairs (Husum Wind 2021; WindEnergy Hamburg 2022), comprehensive trade magazine articles, an upcoming final print brochure, a short public informational film, and the project portal (www.ar4wind.de).
Management Governance: Maintain overall administrative oversight to ensure timely target compliance across the consortium.	EPC gGmbH successfully steered the consortium, ensuring all primary deliverables, technical milestones, and field testing cycles were executed within the designated timeframe. While full software unification of the two final prototype variants remained incomplete due to time and

Predefined Project Target	Actual Achieved Project Outcome
	architectural constraints, all core research goals were successfully fulfilled.

2.2. Results of the Research Project

The collective research and engineering efforts conducted throughout the ar4wind project successfully generated four interconnected pillars of knowledge and software assets:

- **Socio-Political Demand and Potential Analysis Matrix:** Detailed mapping of legal entry points for informal public participation within highly structured statutory planning timelines, alongside technical data mapping the capabilities and constraints of mobile AR tracking architectures.
- **Advanced Geodata-Driven AR Calibration Engineering:** Design and deployment of novel, user-guided spatial registration systems driven by high-resolution 3D digital surface models, backed by algorithmic methods for multi-layered foreground occlusion processing.
- **Deployable Mobile Software Prototypes:** Creation of functional, cross-platform mobile apps for real-time wind park visualization, featuring newly engineered, user-friendly menu systems designed to minimize user friction during field calibration.
- **Evidence-Based Practical Implementation Guidelines:** Synthesis of extensive multi-region field trial insights into a definitive guide for practitioners, detailing optimal group facilitation techniques, process integration methods, and a comparative analysis of real-time ad-hoc AR versus pre-rendered video playback.

2.2.1. Bedarfs- und Potenzialanalyse (Demand and Potential Analysis)

At the project's inception, it was critical to establish a comprehensive understanding of the spatial planning legislation governing wind energy in Germany. The research team systematically mapped the legal procedural phases, statutory consultation deadlines, and stakeholder interaction dynamics across three administrative levels: regional spatial planning (Raumordnung), municipal land-use planning (kommunale Bauleitplanung), and formal immission control approval processes (Immissionsschutzrechtliches Genehmigungsverfahren). This material was published on the project website as a foundational guide. Following this baseline phase, the team gathered empirical feedback from 16 qualitative expert interviews and a technical requirements workshop in November 2021. This combined approach systematically defined the functional requirements for the software interface and identified the core criteria for successful practical deployment in professional environments.

2.2.1.1. Use of AR Visualizations in Public Participation Processes

Public interaction during onshore wind farm planning occurs across three distinct administrative levels, each characterized by unique legal mandates, competent planning authorities, and varying windows of opportunity for public consultation:

Figure 2 & 3: Structural Systematics and Participation Tracks

Wind infrastructure deployment operates within a strict hierarchical multi-level planning system. It spans from state and regional level macro-targeting (Landes- und Regionalplanung) down through municipal micro-zoning (Bauleitplanung: Flächennutzungsplan/Bebauungsplan), concluding at the project-specific building and environmental permitting phase. Within this framework, public engagement splits into mandatory formal tracks dictated by statutory planning codes, and voluntary informal tracks initiated by project developers or municipalities to facilitate early community dialogue.

The characteristics of public engagement vary substantially across these three administrative tiers:

- **Regional Spatial Planning (Regionalplanung):** Public interaction at this macro tier is legally restricted to a single-stage process. The competent regional authority puts fully matured regional plan drafts on public display for comment. Crucially, early statutory notifications at this tier do not constitute open public participation; they function as a unidirectional information broadcast without any collaborative dialogue options.
- **Municipal Land-Use Planning (Bauleitplanverfahren):** This tier utilizes a mandatory two-stage participation structure. The planning municipality must actively engage both public agencies and citizens early in the process when plan concepts are still flexible (Frühzeitige Beteiligung). Later, they must formally expose the finalized plan drafts to allow for official legal objections. Consequently, this tier offers substantially more collaborative opportunities for citizen dialogue than regional macro-zoning.
- **Immission Control Permitting (Genehmigungsverfahren):** Executed under the Federal Immission Control Act (BImSchG), this project-specific phase evaluates compliance with strict environmental criteria, safety mandates, and residential proximity thresholds. Depending on the installation's size, the review follows either a simplified administrative track or a formal track. The formal track mandates a full Environmental Impact Assessment (EIA) and an open public consultation window, which project developers can also opt into voluntarily for smaller installations.

Figures 4, 5 & 6: Statutory Sequences Across Planning Tiers

Official procedural sequences map the step-by-step progress of regional planning, municipal land-use planning, and formal immission control permitting. These charts detail every statutory milestone—from early site analysis and formal drafting resolutions through public exposure windows, objection evaluations, and final regulatory approval. They highlight exactly where informal AR visualization interventions can be injected to optimize public understanding and engagement.

To maximize the utility of mobile AR visualization tools within these complex planning frameworks, the research consortium isolated several high-leverage entry points for informal public dialogue interventions:

- **Early Draft Conception Phase (Regional Planning Level):** During the early stages of spatial drafting, regional planning authorities can deploy mobile AR tools during local informational field tours. This allows them to show affected rural populations exactly how prospective wind energy zones would look with realistic turbine layouts, visually explaining why specific zones are technically preferable.
- **Visibility Assessments for Cultural Heritage Impact Studies (Regional Planning Level):** Regional authorities can leverage the AR visualization engine to conduct real-time field visibility analyses. This supports specialist reviews for state historic preservation offices, helping to evaluate whether projected turbine arrays would visually crowd or degrade listed architectural monuments.
- **Post-Draft Public Exposure Consultation Windows (Regional Planning Level):** During local citizen information workshops following draft plan publications, planners can use the mobile tool to help citizens experience the precise scale and appearance of next-generation turbine heights, anchoring local discussions in accurate spatial data.
- **Early Spatial Master Planning and Concept Drafting (Municipal Land-Use Level):** Municipal civil engineers can use the AR system during early planning phases to brief city council members and zoning boards. This clearly demonstrates how various prospective clean energy overlay zones would interact visually with existing residential boundaries.
- **Zoning Height Limitation Assessments (Municipal Land-Use Level):** Before enacting strict local building height caps, municipal planning departments can run real-time comparative field simulations using the AR tool. This helps expert committees visually assess and debate the differences between various hub heights and rotor dimensions.
- **Early Post-Exposure Citizen Field Excursions (Municipal Land-Use Level):** Municipalities can arrange guided community walking tours within proposed planning zones. Equipped with the AR application, citizens can directly visualize land-use plans in situ, enabling them to submit highly informed and constructive feedback during formal public comment windows.
- **Pre-Permit Submission Consultation Initiatives (Immission Control Permitting Level):** Once land lease agreements are secured and project geometries are fixed, project developers can use AR field tours to proactively present their plans to nearby residents, ensuring clear and transparent communication before submitting formal building permit applications.

- **Environmental Impact Assessment Scoping Formats (Immission Control Permitting Level):** During formal scoping meetings to define the environmental review framework, developers can use the AR visualization tool to enrich Environmental Impact Assessment (EIA) reports with accurate field-validated photo and video logs, providing clear visual evidence for reviewing authorities.
- **Statutory Public Objection Clarification Hearings (Immission Control Permitting Level):** Following formal public comment windows, permitting authorities can use the interactive AR tool during official objection hearings. This helps resolve visual disputes and clarify complex spatial complaints directly with affected stakeholders in real time.

2.2.1.2. Target User Groups of the AR Visualization Tool

Based on insights from 16 detailed expert interviews, the research team mapped the target user ecosystem for the mobile AR application. The analysis identified four distinct primary user groups and two secondary user groups, as summarized below:

User Tier & Group	Core Operational Use Cases & Deployment Contexts
Primary: Municipalities	Acts as a neutral, fact-based information tool for municipal planning departments and local councils. It helps evaluate setback distances, residential buffer zones, and height limits during municipal land-use planning, while ensuring complete transparency during public town hall meetings.
Primary: Regional Planners	Supports regional planning officials during spatial suitability assessments and environmental impact reviews. It allows teams to visualize hypothetical wind farm layouts and evaluate landscape changes before finalizing regional growth plans.
Primary: Process Facilitators	Provides professional moderators and environmental mediators with an interactive tool to prepare for complex community dialogues. It helps neutral facilitators understand project impacts firsthand, and adds an engaging, interactive element that encourages constructive citizen participation during workshops.
Primary: Project Developers	Serves as an internal validation tool for project developers during initial site exploration and field scouting. It also helps capture rich, accurate photo and video assets to support corporate communication and presentations during municipal council sessions.
Secondary: Local Citizens	Enables nearby residents to experience proposed wind projects from their own neighborhoods via neutral, reality-compliant visualizations. This helps protect communities from exaggerated or misleading visual claims from either project proponents or opponents. To ensure an accessible, low-friction experience, this group should access the tool within structured, professionally moderated formats like guided community field tours.
Secondary: Specialist Experts	Assists certified landscape architects and independent environmental experts during the preparation of formal scenic impact assessments and historic visibility reviews, facilitating efficient joint field reviews with regulatory authorities.

Figure 7: User Ecosystem Mapping Matrix

The complete user ecosystem diagram illustrates the functional connections between the primary professional operators (Municipalities, Regional Planning, Project Developers, and Process Support) and the secondary target audiences (Local Citizens and Specialized Environmental Experts), showing how the AR software bridges professional technical workflows and public communication needs.

2.2.1.3. Requirements for Practical Operation within Participation Frameworks

The initial analytical phase isolated a set of mandatory operational parameters that must be fulfilled before institutional actors will choose to adopt and deploy the mAR tool during real-world project planning:

- **Advanced Stage Planning Clarity:** The underlying wind energy proposal must be mature and highly concrete. Deploying the tool without fixed geographical coordinates or defined turbine dimensions causes significant confusion, fuels public misinformation, and leads to unproductive speculative discussions.
- **Pre-Existing Social Alignment:** The planning scenario should ideally be socially stable and uncritical. Utilizing an unvalidated AR app within an ongoing, highly polarized local conflict can weaponize minor technical tracking variances, turning the tool into a target for public distrust.
- **Controlled and Moderated Group Sizes:** The interaction format must take place within a secure, professionally facilitated environment with a manageable audience size (such as small, invitation-only workshops or structured citizen tours). Distributing the tool to large, unmanaged crowds causes immediate technical support bottlenecks that disrupt the educational goals of the event.
- **Integration with Broader Public Processes:** The AR intervention must be explicitly connected to a structured, higher-level public participation timeline. It should function as an integrated milestone within a comprehensive communication strategy, rather than a standalone technology showcase.
- **Presence of Dual Expert Facilitation:** Every deployment must feature active commentary from both a technical software specialist and a professional spatial planner. This ensures user calibration errors are corrected immediately, and planning questions are answered accurately on site.
- **Transparent Communication of Design Limitations:** Facilitators must clearly state the visual and legal boundaries of the presentation. Operators must explicitly communicate that the AR view is a planning tool, not a perfect aesthetic rendering, and clarify which parts of the layout are fixed by law versus which parts remain flexible.
- **High Interface Usability and Intuitiveness:** The application interface must be simple and intuitive. Non-technical users must be able to complete device calibration and view the 3D assets quickly, without requiring extensive training.
- **Low Set-Up Overhead and Cost:** The system must be cost-effective and easy to configure for any given wind project. Project developers and municipalities must be able to import GIS design metrics into the AR environment without requiring expensive, customized software engineering.

2.2.1.4. Functional Requirements for the AR Visualization Tool

During the collaborative design workshops, stakeholders systematically compiled and prioritized a functional requirements catalog for the mobile AR application, categorizing features by technical priority and implementation complexity:

Figure 8: Prioritized Functional Requirements Specifications

The structured functional features index organizes software requirements into three categories: Core Vital Functions (High Priority), Desirable System Enrichments (Medium Priority), and High-Complexity Features with technical feasibility constraints.

1. **Core Vital Functions (High Technical Priority — Implemented):**
    - *Real-Time Landscape Visualization of Planning Variants:* Reality-compliant, position-accurate real-time 3D rendering of multi-turbine wind park configurations superimposed onto live camera views at designated coordinates.
    - *On-Site Parameter Adjustments:* Interactive menus allowing real-time changes to turbine types, tower heights, rotor diameters, blade rotation speeds, and yaw orientations directly in the field.
    - *Dynamic Contextual Information Display:* On-screen text overlays displaying vital spatial metrics, including real-time distance from the viewer to each turbine, global coordinates, and solar positioning vectors.
    - *Ambient Color and Contrast Controls:* Integrated sliders allowing users to manually adjust color hues, brightness values, and contrast levels of the 3D models to match varying weather conditions, seasonal light changes, or evening dusk environments.
    - *Rich Metadata Media Archiving:* One-touch photo and video recording that embeds essential spatial metadata (including exact capture coordinates, time stamps, compass headings, and camera focal lengths) directly into the file attributes to ensure scientific reproducibility.
2. **Desirable System Enrichments (Medium Priority — Implemented in Prototypes):**
    - *Scale Comparison Models:* Integrated options to render familiar virtual structures (such as high-voltage power pylons or well-known municipal buildings) right next to the proposed turbines, helping laypersons accurately comprehend the physical scale.
    - *Planning Boundary Overlays:* Real-time display of designated wind energy suitability zones and local zoning boundaries directly within the AR view to clearly show permissible placement areas.
    - *Existing Infrastructure Information Logs:* Contextual text lookups providing operational metrics for existing wind installations within the viewer's line of sight, such as current power generation data or tower height statistics.

3. **High-Complexity Features (Low Priority — Deferred Due to Technical Overhead):**
    - *Dynamic Shadow Simulation:* Algorithmic rendering of periodic shadow casting across topography based on live astronomical time matrices. Deferred due to massive graphics compute requirements on mobile hardware.
    - *Repowering Scenario Modeling:* Dynamic workflows to selectively hide older existing turbines in the live camera feed and replace them with next-generation models in the same location. Deferred due to the extreme complexity of real-time pixel erasure algorithms.
    - *Night Marking Aviation Simulations:* Real-time dark mode simulation that mimics nighttime environments and activates synchronized flashing red aviation safety lights on the virtual turbine nacelles. Deferred due to limitations in real-time camera contrast adjustments at night.
    - *Dynamic Weather Atmospheric Effects:* Automated rendering modifiers that adjust visibility based on real-time fog layers, cloud cover densities, or rain patterns. Deferred due to high processing overhead.

2.2.2. GeoAR Localization and Calibration (GeoAR Registration)

The stable, accurate rendering of virtual objects placed kilometers away at explicit coordinates is termed Geodata-Based Augmented Reality (GeoAR). While indoor AR frameworks can rely entirely on close-range surface tracking (e.g., Google ARCore, Apple ARKit), large-scale outdoor GeoAR requires overcoming unique computer vision and spatial tracking challenges. Currently, consumer mobile operating systems do not provide native tools to resolve these outdoor tracking issues.

Figure 9: Foundational Visual Tracking Failures in Uncalibrated Outdoor AR

Uncalibrated outdoor AR systems suffer from two primary tracking failures: 1) Spatial Localization Errors caused by sensor inaccuracies that misalign virtual assets and distort public perception of turbine placement, and 2) Occlusion Rendering Failures where the lack of environmental depth awareness causes distant turbines to erroneously render in front of near-field trees or buildings, destroying visual realism.

2.2.2.1. Technical Challenges of Geodata-Based Augmented Reality

The research consortium focused on resolving two core technical tracking challenges to ensure reliable performance in outdoor environments:

- **High-Precision Spatial Geo-Localization:** Achieving stable, position-accurate AR rendering requires establishing the mobile device's exact 3D location and orientation vectors within a global geographic coordinate system (GeoAR Localization). If the orientation metrics are off by even a fraction of a degree, the virtual turbine will drift significantly on screen, leading to an incorrect perception of its physical location in the landscape. Standard built-in consumer mobile sensors (consumer GPS chips and digital compasses) are highly susceptible to electromagnetic interference and signal degradation, meaning alternative calibration methods are required to ensure tracking stability.
- **Reality-Compliant Multi-Layered Occlusion Processing:** To maintain visual realism, distant virtual objects must be correctly obscured by real foreground elements like trees, buildings, or local hills. If an AR engine lacks awareness of environmental depth, a turbine located three kilometers away will render on top of a tree line just fifty meters away, ruining the illusion of depth. Because standard mobile depth sensors only map objects within a few meters of the device, developers must engineer alternative depth mapping systems to handle far-field occlusions across expansive regional landscapes.

Resolving these tracking challenges requires a robust initialization process (GeoAR Calibration). Before displaying any virtual assets, the mobile device must be precisely calibrated on site to align its virtual camera parameters with the physical geography. While automated computer vision methods that match camera views to pre-computed 3D point clouds exist, they require massive data pipelines and are expensive to scale across broad rural landscapes. Consequently, the ar4wind project focused on engineering and validating three user-controlled manual calibration methodologies.

2.2.2.2. Calibration Method A: Reference Point Alignment and Color-Based Sky Segmentation

This tracking methodology utilizes manual reference point alignment combined with automated color thresholding to handle environmental occlusions, operating through three sequential user steps:

Figure 10: Operational Workflow for Calibration Method A

The user workflow for Method A proceeds through three steps: 1) Manual position adjustment on a high-resolution satellite map view, 2) Orientation alignment by targeting a prominent real-world landmark with a virtual crosshair, and 3) Selection of sky colors to generate an automated foreground occlusion mask.

1. **Map-Based Position Initialization:** The system captures an initial location estimate using the device's internal GPS. The user then views a high-resolution satellite map and manually drags a crosshair to correct any sensor positioning errors, establishing an accurate baseline location.
2. **Visual Orientation Alignment:** To calibrate the device's heading and compass vectors, the user selects a prominent, easily identifiable real-world landmark (such as a church steeple or a distinct rooftop) on the map interface. The app then displays a vertical virtual alignment line over the live camera view. The user rotates the physical device until the crosshair matches the physical landmark, instantly correcting any internal compass drift.
3. **Color-Based Sky Segmentation Filtering:** To handle occlusions without loading heavy 3D terrain data, the system uses real-time color filtering. The user taps a typical patch of sky on the screen, prompting the software to generate a dynamic color mask that isolates the sky area. The rendering engine treats the masked sky as empty space where virtual turbines can render freely, while any non-matching foreground items (like trees or buildings) automatically act as occlusion barriers that block the virtual assets.

The primary advantage of Method A is its exceptional geographic flexibility; it can be deployed instantly anywhere in the world without requiring expansive pre-loaded 3D structural data. However, it requires a time-consuming setup process with multiple manual alignment steps. Users sometimes struggle to locate matching landmarks between the 2D map view and the physical landscape. Furthermore, the color-based sky segmentation filter performance degrades significantly under challenging weather conditions, such as high-contrast blue skies filled with bright white clouds.

2.2.2.3. Calibration Method B: Virtual 3D Landscape Model Alignment

This methodology utilizes public high-resolution 3D geographic data to calibrate the camera and handle environmental occlusions. The system relies on an offline geodata processing pipeline that prepares raw geographic assets for mobile use:

Figure 11: Optimized Geodata Processing Pipeline Architectural Schema

The processing pipeline converts raw, text-heavy GIS source datasets (including Digital Surface Models, Digital Terrain Models, and CityGML files) into lightweight, high-performance 3D spatial tiles optimized for real-time rendering on mobile hardware.

The automated geodata pipeline processes public spatial datasets through five core optimization steps:

- **Coordinate Reference System Conversion:** Re-projects diverse source geodata matrices into a single, unified coordinate reference frame used by the mobile client app.
- **Spatial Grid Tiling (Tiling):** Subdivides massive regional datasets into small, localized geographic tiles that can be streamed efficiently on demand based on the user's location.
- **Mesh Simplification and Compression:** Applies advanced poly-reduction algorithms to shrink file sizes while preserving important visual features and terrain profiles.
- **Optimized 3D Mesh Generation:** Converts optimized geographic coordinates into compact, high-performance 3D model files suitable for mobile graphics processors.
- **Metadata Attribute Linking:** Pairs each data tile with structural metadata attributes, saving the assets to a central web server that serves data to mobile devices on request.

Once the localized 3D terrain models are loaded into the mobile client, the on-site user calibration sequence follows a simple two-step workflow:

Figure 12: Operational Workflow for Calibration Method B

The user workflow for Method B consists of: 1) Selecting the user's current position on a satellite map view, and 2) Aligning a translucent 3D terrain overlay with the physical landscape using intuitive multi-touch swipe and pinch gestures.

1. **Map-Based Position Selection:** The user refines their physical location on a standard map interface to load the corresponding 3D terrain tile from the server.
2. **Interactive Multi-Touch Mesh Alignment:** The app displays a translucent 3D overlay of the local terrain features over the live camera view. Because internal compass sensors are often inaccurate, this overlay will initially appear misaligned. The user uses simple touch gestures to adjust the overlay: a single-finger horizontal swipe rotates the 3D model to align with physical hills or buildings, while a two-

finger vertical swipe adjusts the virtual camera height. Once the translucent model matches the physical scenery, the system is calibrated. The aligned 3D model is then made invisible, functioning as a high-precision depth barrier that automatically handles far-field occlusions.

Figure 13: Empirical Evaluation and Operational Best Practices for Method B

The operational guidelines matrix highlights key deployment practices for Method B: it delivers highly accurate device orientation (within 1.0 degree of accuracy), benefits from clear interactive tutorials, works reliably across diverse landscapes, and performs best on large smartphone displays used in portrait mode.

The primary advantage of Method B is its speed and user-friendly interaction loop, allowing users to complete calibration within seconds. Because it utilizes real 3D surface data, the system handles complex far-field occlusions automatically, eliminating the need for manual sky segmentation. However, the system requires significant server infrastructure and data bandwidth to stream large 3D files. Furthermore, high-quality, up-to-date public 3D geodata models are not yet uniformly available for free across all administrative regions in Germany.

2.2.2.4. Calibration Method C: Pre-Registered Spatial Anchors

To eliminate complex manual calibration steps for end-users during public events, the team designed a two-tier workflow utilizing pre-registered spatial anchors:

Figure 14: Automated Camera Calibration via Pre-Registered Spatial Anchors

The automated Method C workflow splits into two parts: 1) An advanced expert setup phase where a specialist pre-registers a high-resolution 3D point cloud anchor at a specific site, and 2) A simplified end-user experience where citizens scan the environment to achieve instant, automated calibration.

1. **Expert Pre-Registration Phase:** A trained technical specialist visits designated viewing locations prior to a public event. Using a specialized expert version of the software, the technician calibrates the device using the 3D landscape method (Method B). They then scan the local environment with the device camera to capture a detailed, high-resolution 3D point cloud of local landmark features (such as signs, rocks, or trees), creating a georeferenced spatial anchor that is saved to a central server.
2. **Automated Public User Onboarding:** During a public participation event, regular citizens open the standard client app at the designated viewing locations. The software automatically downloads the pre-registered spatial anchor based on the device's GPS position. The user is prompted to gently scan the local environment with their camera for a few seconds. Once the software matches the live camera view to the saved 3D point cloud anchor, the system achieves instant, automated spatial calibration. The virtual wind turbines and 3D occlusion layers render immediately with high precision.

The primary benefit of Method C is the complete removal of complex manual calibration tasks for regular users, making the application accessible to audiences with no technical background. The main limitation is that visualizations are restricted to pre-registered viewing locations. Additional research is also required to assess how seasonal changes (such as winter snow cover or autumn leaf loss) affect the reliability of the saved 3D point cloud anchors.

While Calibration Method A and Method B were fully implemented and validated during public trials, Method C was developed as a functional proof-of-concept prototype that requires further field testing in future research cycles.

2.2.3. Mobile Prototype Demonstrator Applications

The consortium engineered three distinct mobile prototype applications to support iterative field testing and technical validation:

- **Application Prototype 1:** Built upon foundational code provided by LandPlan OS, this initial version utilized Calibration Method A (manual reference points). It served as the primary testing asset during Phase I public trials to gather initial feedback on general usability and features.

- **Application Prototype 2A:** Developed as an optimized iOS application for Apple iPad, this version featured a completely redesigned user interface that streamlined the setup wizard and improved the controls for Method A (manual reference points and sky segmentation).
- **Application Prototype 2B:** Developed as an optimized Android application for the Samsung Galaxy Tab, this version focused on validating Calibration Method B, successfully integrating real-time 3D landscape data and gesture-based alignment into a smooth user workflow.

Due to fundamental architectural differences in the software backends and data structure layouts between the systems, merging both calibration variants into a single unified application was not feasible within the remaining project timeline. Both versions utilize the Unity engine as their core graphics framework, ensuring robust real-time rendering performance.

Figure 15: Structural Screen Captures from Application Prototypes 1, 2A, and 2B

Comparative interface captures illustrate the visual evolution of the software: from the complex layout of Prototype 1 to the refined, wizard-guided menu systems featured in the optimized versions 2A and 2B.

2.2.3.1. User Interface Design Concept

The user interaction concept for the optimized version 2 applications follows a highly structured, sequential three-stage workflow to minimize user confusion:

Figures 16 & 17: User Interface Wireframes and Layouts

Comprehensive user interface schemas document the complete functional architecture of the software: mapping from the initial project selection screen and sequential calibration steps (S0–KA4) through to the main AR visualization view, property modification menus, and map views (S1–S6).

The interface layout contains seven core functional screens, structured as follows:

- **Screen S0 (Project Selection Dashboard):** The application launch screen displays a clean overview of all available wind farm simulation scenarios configured for the region. A prominent download button pulls all required assets (such as specific 3D turbine models, spatial coordinates, and local terrain tiles) from the server into local storage. Once the download is complete, a clear start button initializes the simulation.
- **Screen KA-1 / KB-1 (Map-Based Position Selection):** The initial step in the calibration sequence displays a full-screen satellite map. The interface places a central target crosshair over the user's current location based on the device's GPS data. The user uses standard drag and pinch gestures to align the crosshair with their exact physical position, correcting any baseline GPS inaccuracies. A top-mounted progress bar continuously tracks the user's position within the calibration sequence.
- **Screen KA-2 & KA-3 (Method A Landmark Alignment Windows):** Used exclusively in Method A, this sequence requires the user to select an easily identifiable physical landmark (such as a church tower) on the map interface (KA-2). The app then switches to a live camera view featuring a prominent vertical alignment line (KA-3). The user rotates the physical device until the virtual line matches the physical landmark, instantly calibrating the internal heading and compass sensors.
- **Screen KA-4 (Method A Sky Segmentation Interface):** This interface displays a movable bounding box over the live camera view. The user positions the box over a clear section of sky and taps a "Capture Color" button to add the selection to an active color filtering list. The app overlays a translucent green mask over the isolated sky areas in real time to show the active occlusion area, and includes an adjustable slider to refine the filter's color sensitivity.
- **Screen KB-2 (Method B Gesture Alignment Interface):** Used exclusively in Method B, this interface loads a translucent 3D model of local terrain features directly over the live camera view. Simple single-finger touch gestures rotate the 3D model horizontally to match physical landscape lines, while two-finger vertical gestures adjust the virtual camera height to complete calibration within seconds.
- **Screen S1 & S2 (Main AR Visualization and Capture Views):** The primary simulation interface displays the live camera view with high-precision 3D wind turbine models rendered seamlessly at their exact geographic positions (S1). A camera icon triggers an instant high-resolution image capture (S2) that saves the view directly to the device, displaying a temporary notification overlay with the saved file name.

- **Screen S3 & S4 (Turbine Inventory and Parameter Modification Panels):** Tapping the inventory icon opens a clean side panel listing all turbines in the active scenario, detailing their operational status, model names, and real-time distances from the viewer (S3). Selecting an individual turbine highlights it on screen while lowering the opacity of adjacent models. Tapping the "Modify" icon opens a property panel (S4) featuring drop-down menus to change turbine models and interactive sliders to adjust hub heights or rotor orientations.
- **Screen S5 & S6 (Environment Settings and Split Map Overlays):** Tapping the settings icon opens a panel to adjust visual rendering properties, including ambient lighting controls and color filters to optimize visibility during low-light evening conditions (S5). It also includes a manual height override slider for viewing simulations from elevated positions like balconies. Tapping the map icon opens a split-screen layout (S6) that shows the user's location and viewing angle relative to the entire wind park installation on a live map.

2.2.3.2. Technical Software Implementation

All application prototypes were built using the Unity 3D engine, providing a high-performance, cross-platform graphics pipeline suitable for deploying on both iOS and Android mobile devices from a shared code baseline. Prototype 1 and Prototype 2A were compiled and validated on Apple iPad systems, while Prototype 2B was deployed and tested on Samsung Galaxy Tab Android platforms.

Figure 18: Integrated Development Environment Platform View

The development environment screenshot illustrates the core software components within the Unity project editor, highlighting the spatial tracking modules, asset management structures, and custom C# rendering scripts used to build the application.

The application software architecture relies on five interconnected core components:

- **3D Graphics Rendering Engine:** Manages the loading of highly detailed 3D turbine assets, handles real-time animations, and processes multi-layered foreground occlusion shaders.
- **User Interface and Interaction Layer:** Manages touch inputs, coordinates menu animations, and handles interactive map elements.
- **Core AR Tracking Framework:** Leverages Unity AR Foundation to interface directly with native device tracking arrays (Apple ARKit and Google ARCore), managing real-time camera feeds, device tracking data, and spatial coordinate anchoring.
- **Spatial Geodata Processing Module:** Manages the ingestion and parsing of complex geographic data, handles coordinate transformations, and processes terrain height matrices.
- **Scenario Configuration Module:** Coordinates data transfers with the web server, manages local asset caching, and handles user scenario configurations.

To support advanced spatial processing, the development team integrated several specialized open-source software libraries: *JSON.NET* for structured data handling, *LibTiff* for reading high-resolution GeoTIFF elevation files, *SharpMap* for managing geographic coordinate transformations, *InAppBrowser* for embedding responsive web elements, and the industrial-grade *GDAL Library* and *MeshLab* engines for automated 3D mesh optimization. Prototype 1 and Prototype 2A connect to a specialized web server backend managed by LandPlan OS to load project configurations, coordinate data, and spatial maps. Conversely, Prototype 2B stores all 3D geodata and scenario assets locally on the device, creating an architectural difference that presented a significant barrier to merging both approaches within the project timeline.

2.2.4. Practical Testing and Evaluation

The consortium conducted two comprehensive field trial phases across diverse municipal regions to evaluate the software's performance and practical utility under authentic planning conditions:

Trial Evaluation Phase	Geographic Locations & Date Parameters	Methodological Evaluation Approach & Target Audiences
Practical Test Series I	Oederan (SN) — April 30, 2022 Freigericht (HE) — May 07, 2022 Bornheim (NRW) — May 19, 2022 Gehrden (NI) — June 02, 2022	Utilized Prototype 1 to gather broad feedback on general feature utility. Collected quantitative survey responses from approximately 100 participants, backed by 10 detailed follow-up telephone interviews. Target cohorts were primarily composed of local elected officials and residents directly affected by upcoming wind projects.
Practical Test Series II	Bruchsal (BW) — February 28, 2023 Freigericht/Alzenau (HE/BY) — March 21, 2023 Berlin Wuhlheide (BE) — April 24, 2023	Utilized Prototype 1 alongside the optimized version 2 applications. Relied entirely on qualitative participant observation methods to analyze process integration, compare ad-hoc field AR against pre-rendered video playback, and evaluate the redesigned user interface. Target cohorts consisted of highly targeted stakeholder groups, local authorities, and expert facilitators.

Figure 19: Field Evaluation Trials Inside Authentic Planning Jurisdictions

Field photographs document the public testing phases: showing municipal stakeholders participating in a guided field tour in Freigericht using Prototype 1 (left), and a specialized comparative research trial in the Berlin Wuhlheide forest using Prototype 2B (right).

2.2.4.1. Practical Test Series I (Baseline Prototypic Evaluation Loops)

The first field trial phase engaged approximately 100 participants. Except for the open public session in Oederan, organizers intentionally invited local elected officials and community representatives who were actively participating in local zoning decisions. A total of 48 participants completed comprehensive evaluation surveys. The demographic breakdown showed a predominantly male cohort with very little prior experience using Augmented Reality applications. Age distribution was evenly split between two brackets: 30–45 years old and 45–65 years old. The majority of participants identified as residents directly affected by local wind proposals who did not have any existing wind turbines near their homes, and over half held active local political offices.

The evaluation focused on tracking system usability and general utility—specifically assessing whether the application could render wind turbines realistically regarding scale, color, and geographic placement. Statistical analysis indicated that prior AR experience, existing familiarity with wind energy infrastructure, or holding political office had no significant impact on survey responses, with all user groups providing highly consistent feedback. The vast majority of participants gave highly positive ratings to both the application's general utility and its visual realism.

Crucially, because the first prototype version presented significant technical calibration hurdles, participants were unable to test secondary application features during the public field sessions. Most of the event time was spent helping users complete the complex manual calibration process (Method A). While most users successfully completed calibration with hands-on assistance from technical staff and reported it as a deeply positive experience, no time remained to test or evaluate advanced features like turbine type modification, photo archiving, or color adjustments. Consequently, the survey data showed high drop-out rates for questions regarding secondary features. The primary insight from Series I was that while the initial prototype functioned effectively as a specialized tool for planning professionals and project developers, it presented an excessively steep learning curve for regular citizens. This finding highlighted the urgent need to develop a highly intuitive user interface and a streamlined calibration process for non-technical users.

From an operational standpoint, the evaluation isolated four critical factors that dictate whether the visualization tool can be successfully adopted for local public engagement:

- **Stability of Local Stakeholder Networks:** Successful deployment requires clear, established communication lines and trusted networks between project developers, local authorities, and the community.

- **Seamless Integration into Public Timelines:** The tool must be explicitly embedded as a core element within an established municipal participation framework, rather than used as a standalone technology showcase.
- **Proactive Commitment from Local Leaders:** Direct, enthusiastic engagement and leadership from municipal authorities and mayors significantly enhances public participation and trust.
- **Careful Pre-Selection of Viewing Sites:** The physical layout of chosen viewing locations must be carefully evaluated beforehand to ensure clear sightlines and minimal sensor interference.

During local feedback sessions, several planning experts questioned whether simpler, pre-rendered AR video logs captured from designated viewing spots could deliver the same communicative benefits as real-time ad-hoc AR visualizations, while eliminating technical configuration friction for users. The research team integrated this comparative question directly into the design of the second field trial series.

2.2.4.2. Practical Test Series II (Process Fit and Comparative Analysis)

The second field trial phase engaged approximately 30 participants across highly focused stakeholder groups. The final specialized evaluation in the Berlin Wuhlheide forest was conducted with a cohort of eight select participants. These trials focused on evaluating process integration—specifically analyzing how professional facilitators can best deploy the tool to foster transparent, constructive community dialogue—and assessing the usability of the newly redesigned version 2 user interface.

The field sessions in Bruchsal and Freigericht/Alzenau were initiated directly by local municipal leaders (including mayors and local climate managers) to support ongoing planning consultations. The Bruchsal deployment featured a structured bus tour for city council members to evaluate prospective wind zones. In contrast, the Freigericht/Alzenau tour focused on capturing high-quality photo and video assets to support an upcoming inter-municipal steering committee session, involving a small team of two operators. These trials revealed that organizing regional field events requires significant logistical coordination. Planners frequently encountered high administrative friction when trying to align technical spatial coordinates between GIS software packages, consumer mapping tools, and the AR application data formats. To ensure smooth execution, organizers must prepare comprehensive print maps for participants that clearly detail proposed turbine locations, viewing stations, setback distances, and the exact tour route.

The final field trial in the Berlin Wuhlheide forest utilized a realistic mock planning scenario, featuring twelve virtual turbines placed between 1,000 and 2,000 meters away from designated viewing locations. Participants were split into two separate testing groups to conduct a structured comparative analysis:

- **Group 1 (Pre-Rendered Video Tracking Track):** This cohort visited designated field stations equipped with standard mobile devices to view pre-rendered AR video logs. Users scanned a physical QR code at each station to open a 20-second high-resolution YouTube video showing a smooth 180-degree horizontal camera pan of the landscape with the virtual wind farm perfectly integrated. These video assets were recorded beforehand by technical specialists using the advanced AR system.
- **Group 2 (Real-Time Ad-Hoc AR Track):** This cohort visited the same viewing stations but used the newly optimized Prototype 2B application to independently configure and experience real-time ad-hoc AR visualizations, using the 3D landscape matching method (Method B) for device calibration.

The field trial yielded overwhelmingly positive feedback for the redesigned version 2 user interface. Participants reported that the new step-by-step sequential calibration wizard and clean menu layout drastically reduced confusion, allowing non-technical users to complete calibration quickly and independently. The trial produced a comprehensive comparative performance matrix detailing the strengths and limitations of both visualization formats:

Comparative Feature Performance Matrix

- Completely eliminates visibility issues caused by direct sunlight or screen glare in the field.

Pre-Rendered AR Video Playback (Group 1)	Real-Time Ad-Hoc AR Visualization (Group 2)
Operational Advantages: • Provides an easy, stress-free user experience by eliminating technical calibration tasks for the public. • Allows participants to maintain a comfortable posture, viewing the video while relaxed or seated without needing to hold a device up continuously. • Can be viewed on any standard consumer device without requiring specialized hardware or on-site software installation.	**Operational Advantages:** • Significantly enhances geographic orientation, helping users easily understand exactly where they are standing and which direction they are looking. • Drastically reduces feelings of skepticism or external manipulation, as users control the device, choose their own viewing angles, and pan at their own pace. • Significantly increases public trust and perceived credibility of the data, as the transparency of the real-time 3D depth alignment makes the visualization feel authentic. • Delivers a deeper sense of scale and realism, making the proposed infrastructure feel tangible and true to life.

Pre-Rendered AR Video Playback (Group 1)	Real-Time Ad-Hoc AR Visualization (Group 2)
Operational Limitations: • Creates a passive experience that can feel scripted or artificial, increasing psychological distance from the planning content. • Restricts the user's view entirely to the pre-recorded camera path, preventing independent exploration. • Fails to communicate occlusion dynamics clearly, as users cannot see the underlying spatial models that block turbines behind trees.	**Operational Limitations:** • Requires a brief user onboarding process and tech introduction, as the interface is not completely self-explanatory. • Requires significant physical effort, as users must hold mobile devices up at eye level for extended periods. • Can be disrupted by extreme weather conditions, such as strong direct sunlight that causes severe screen glare.

The comparative analysis demonstrates that both presentation formats possess distinct communicative values and can be deployed effectively depending on the target audience and context. Organizers can choose to deploy low-friction pre-rendered videos for broad public distribution, or utilize the interactive ad-hoc AR application for focused, moderated stakeholder workshops. Furthermore, the usability of pre-rendered videos can be significantly improved by using professional camera stabilizers, adding clear audio commentary, highlighting occluded turbines with color markers, and starting clips with static grounding frames to help viewers anchor themselves in the landscape.

2.2.4.3. Recommendations for Action for Practical Operation

By analyzing the data across all field trials, the research consortium synthesized a definitive set of actionable guidelines for utilizing mobile AR tools inside public participation frameworks:

- **Ensure Purposeful Process Integration:** The visualization tool must be explicitly integrated into a secure, structured public participation framework. Its deployment requires careful advance planning, including site scouting, route mapping, pre-configuring test hardware, and securing fallback devices.
- **Prioritize Mature and Uncritical Planning Cases:** The software should ideally be deployed for projects with well-defined parameters that have already been introduced to local councils and residents. If a project's parameters are still flexible, planners must pre-configure multiple distinct scenario variants in the app and allocate extra time during events to explain the choices, keeping local discussions focused on constructive options.
- **Mandate Planning and Technical Facilitation:** AR sessions must be accompanied by qualified spatial planners who can provide professional commentary and answer zoning questions. The tool should not be distributed to unassisted individuals; it performs best during structured, guided field tours that use printed maps to complement the digital interface.
- **Incorporate Structured Technical Onboarding:** Because the device calibration process is not entirely self-explanatory, events must begin with a brief, highly structured technical introduction to ensure users complete calibration accurately and experience a stable simulation.

- **Leverage Ad-Hoc AR to Foster Trust:** Real-time ad-hoc visualizations should be prioritized during key stakeholder meetings. The interactive nature of the live 3D terrain matching increases transparency and builds significant public trust in the validity of the data compared to static images.
- **Deploy Pre-Rendered Videos for Broad Outreaches:** For large public events or general online distribution, organizations should utilize pre-rendered AR video logs. These assets eliminate technical troubleshooting, run on standard consumer devices, and avoid screen visibility issues on bright, sunny days.
- **Streamline Project Configuration Pipelines:** Project developers must be provided with simple, intuitive web portals to easily upload standard GIS layout files and coordinate metrics into the AR platform, minimizing the technical expertise needed to set up new project simulations.
- **Establish Regional Technology Services:** Municipalities should be able to partner with regional energy agencies, mobile dialogue platforms, or public mediation offices that maintain inventories of pre-configured tablets and provide trained staff to facilitate local events smoothly.

2.3. Expected Benefits and Exploitation of Results

The ar4wind research initiative successfully demonstrated that mobile Augmented Reality technology can effectively anchor polarizing clean energy space disputes onto objective facts. Beyond advancing technical software architectures, the project's primary value lies in defining the exact operational criteria and practical guidelines needed to successfully integrate immersive tech into modern public planning routines. The core utility of the project's outcomes focuses on four high-leverage domains:

- **Highly Accurate, Credible On-Site Visualizations:** Extensive field trial metrics confirmed that the application's real-time 3D rendering delivers exceptional visual accuracy and data credibility. By allowing stakeholders to experience proposed infrastructure projects from authentic local perspectives, the system provides an invaluable tool for evaluating landscape modifications and scenic impacts, far outperforming traditional static photomontages or pre-rendered videos. The updated, user-friendly interface designs successfully resolve previous usability issues, providing an efficient tool ready for professional planning environments.
- **Clear Identification of Technical Requirements and Needs:** The initial analytical phases produced a definitive matrix of the technical requirements, hardware constraints, and user needs associated with deploying AR tools for wind energy planning. These documentation assets provide a valuable blueprint for software architects and public agencies looking to engineer immersive visualization systems for infrastructure projects.
- **Evidence-Based Guides for Public Engagement:** Multi-region field testing provided deep insights into user behavior and process fit, which were synthesized into concrete recommendations for action. This guide helps public mediators and municipal planners minimize technical friction, select optimal interaction formats, and design effective communication tracks during local clean energy transitions.
- **Robust and Stable GeoAR Spatial Calibration Frameworks:** The project successfully engineered a novel outdoor calibration methodology utilizing high-resolution 3D digital surface models to achieve stable device registration in diverse environments. This framework represents a significant advancement in computer vision for outdoor tracking, laying a solid foundation for future automation, such as the integration of automated image recognition algorithms or hands-free AR smart glasses.

The consortium designed the software architecture with flexible interfaces, allowing the technology to be easily adapted to a wide range of public infrastructure projects beyond wind energy. The core visualization and calibration systems can be readily configured to simulate utility-scale solar installations, high-voltage transmission lines, or urban greening initiatives like public space transformations and building facade retrofits.

Mandatory Technical Requirements for Commercial Deployment

To transition the developed prototypes into a fully commercial, market-ready platform, developers must execute further optimization cycles. Future work must focus on expanding automated error-handling routines, enhancing application stability under extreme thermal conditions, adding integrated offline operational modes, and engineering scalable data pipelines to streamline the ingestion of public geodata matrices across varying administrative regions.

2.4. Field Progress at Other Entities

Engineering stable outdoor mAR systems that render far-field virtual assets accurately across expansive regional geographies requires resolving complex spatial registration challenges. Alongside the user-guided manual calibration systems engineered during this project, major technology corporations have recently focused on developing automated, cloud-based visual positioning networks.

The most notable development in this domain is Google's *Visual Positioning System (VPS)*, made available via the *ARCore Geospatial API*. The Google VPS engine utilizes advanced computer vision to match live camera frames against an expansive, pre-computed global database of 3D point clouds extracted from Street View imagery, enabling instant, automated spatial tracking. However, this framework relies on proprietary data structures and depends entirely on the availability and accuracy of Google's pre-existing street logs. Consequently, the automated system is strictly limited to paved public roads and urban environments. It fails completely when deployed in rural areas, private properties, or natural landscapes (such as open agricultural fields, dense woodlands, or rolling hills)—the exact locations where onshore wind farms are typically planned. Thus, the manual geodata-driven calibration methods developed for the ar4wind application remain essential for regional infrastructure planning, providing complete operational independence from external proprietary tracking networks.

Within the commercial clean energy sector, mobile AR applications remain exceedingly rare due to these tracking challenges. The authors are aware of only two comparable prototype applications currently in development: an AR visualization tool engineered by *BayWa r.e.* configured exclusively for a single wind farm project (Langenbrander Höhe), and the *ReViSAR* software application engineered by utility provider *EnBW* to visualize clean energy installations. However, public documentation regarding whether these corporate applications are integrated into structured community dialogue frameworks is completely unavailable. Within current academic literature, the ar4wind project represents the only comprehensive study analyzing the technical performance, public impact, and process integration requirements of mobile AR tools within public participation frameworks.

2.5. Publications and Technical Presentations of the Project Consortium

The project placed a high priority on continuous knowledge transfer and public dissemination throughout its timeline. The following sections catalog the academic papers, technical lectures, public relations campaigns, and workshops delivered by the consortium partners:

2.5.1. Academic and Technical Publications

- Burkard, S., Poppel, M., Deharde, M., & Fuchs-Kittowski, F. (2024). *Georeferenced Anchors for Localization in Outdoor Augmented Reality Applications - Development of a Framework for Automated Global Registration in Geodata-Based AR Scenarios*. In F. Fuchs-Kittowski, A. Abecker, F. Hosenfeld, A. Reineke, & C. Jolk (Eds.), Environmental Information Systems - Digitalization in the Context of Climate Change and the Energy Transition. Wiesbaden: Springer Vieweg (Forthcoming).
- Burkard, S., Fuchs-Kittowski, F., Deharde, M., & Poppel, M. (2023). *Spatial Anchors for Mobile Environmental Tracking Pipelines*. Compiled for the 30th Annual Symposium of the Corporate Environmental Information Systems Working Group, Springer-Verlag.
- Burkard, S., & Fuchs-Kittowski, F. (2022). *Design and Development of a Mobile Outdoor AR Application for On-Site Visualization of Wind Turbines*. In Proceedings of the 7th International Congress on Information and Communication Technology (ICICT 2022), Lecture Notes in Networks and Systems, Vol 447. Singapore: Springer, pp. 101-111. https://doi.org/10.1007/978-981-19-1607-6_27
- Burkard, S., & Fuchs-Kittowski, F. (2022). *Mobile Outdoor AR Application for Precise Visualization of Wind Turbines Using Digital Surface Models*. In Proceedings of the 8th International Conference on Geographical Information Systems Theory, Applications and Management (GISTAM 2022), Setúbal, Portugal: Scitepress, pp. 15-24. https://doi.org/10.5220/0010989600003185
- Burkard, S., Fuchs-Kittowski, F., Deharde, M., Poppel, M., & Schreiber, S. (2022). *A Mobile Augmented Reality Application for Rendering Planned Wind Power Facilities*. In F. Fuchs-Kittowski & A. Abecker (Eds.), Environmental Information Systems - How Digitalization Contributes to Sustainability. Wiesbaden: Springer Vieweg. https://doi.org/10.1007/978-3-658-35685-9_2
- FA Wind. (2022). *Institutional Participation Frameworks inside Regional and Municipal Zoning Sequences*. Industry Briefing Paper, Berlin.

2.5.2. Academic Lectures and Presentations

- **March 13, 2023:** Technical Presentation delivered at HTW Berlin: *"Augmented Reality for On-Site Visualizations of Renewable Energy Projects in Participation Processes"*, delivered during an international expert exchange session with researchers from the Institute for Sustainable Energy Policies of Nagoya University, Japan.
- **April 25, 2023:** Online Presentation delivered at the FA Wind Board of Directors Session: *"AR Visualizations of Wind Facilities – Opportunities and Constraints for Public Communication"*.
- **May 11, 2023:** Online Lecture delivered at the National UIS Workshop 2023: *"Georeferenced Anchors for Localization in Outdoor Augmented Reality Applications - A Framework for Automated Global Registration"*.

- **June 06, 2023:** Keynote Lecture delivered at the Central German Renewable Energy Industry Day in Halle (Saale): *"On-Site Visualization of Projected Wind Installations Using Augmented Reality - Strengths and Operational Limits for Community Engagement"*.
- **June 17, 2023:** Public Demonstration and Interactive Poster Session presented at the Long Night of Sciences (Lange Nacht der Wissenschaften) in Berlin: *"Visualizing the National Energy Transition Through Mobile Augmented Reality"*.
- **February 24, 2022:** Technical Session Presentation delivered at ICICT 2022: *"Design and Development of a Mobile Outdoor AR Application for On-Site Visualization of Wind Turbines"*.
- **April 28, 2022:** Technical Session Presentation delivered at GISTAM 2022: *"Mobile Outdoor AR Application for Precise Visualization of Wind Turbines Using Digital Surface Models"*.
- **September 06, 2022:** Scientific Lecture delivered at the GeoForum MV Conference in Rostock: *"Geodata-Driven Calibration Frameworks for Outdoor Augmented Reality – High-Precision On-Site Overlay via Integrated 3D Terrain Meshes"*.
- **October 18, 2022:** Interactive Research Poster Session presented at the HTW Berlin Research Forum: *"Mobile Outdoor Extended Reality Environments for Citizen Participation Engineering"*.

2.5.3. Public Relations and DisseminDissemination Campaigns

- **September 06, 2022:** Extended media interview with Michael Hahn for the feature article *"Seeing Wind Farms Before Construction Starts"*, published in *Neue Energie* Magazine, Issue No. 10/2022, the National Magazine for Climate Protection and Renewable Energies.
- **March 04, 2023:** Media interview with Lars Klaaßen regarding *"Structural Challenges in Public Engagement for Wind Energy Proposals using Immersive Tech"*, published in the official BWE Annual Yearbook 2023.
- **March 09, 2023:** Technical interview with Lars Klaaßen regarding *"Algorithmic Tracking Potentials and Constraints in Geodata-Driven Mobile AR"*, published in the official BWE Annual Yearbook 2023.
- **June 2023:** Publication of the digital project results print brochure and production of a short informational documentary film showcasing the final software prototypes.

Structured Knowledge Workshops

- **November 24, 2021 (Online Requirements Workshop):** *"Deploying mAR Technology within Wind Infrastructure Operations"*. Featured technical presentations on the limits of mobile tracking, followed by interactive digital whiteboarding sessions with 20 key stakeholders to map user groups and functional workflows.
- **January 20, 2023 (Dialogue Expert Workshop):** *"Wind in the Viewfinder 2.0 – Environmental Mediation and AR Visualization Integration"*. Featured introductory lectures on user requirements, followed by structured focus group debates structured after the World Café format to optimize public engagement practices.

General Public Relations Measures

- **Web Portal Launch:** Designed, deployed, and maintained the central public project website (https://www.ar4wind.de/).

- **Communication Assets:** Designed and distributed a unified project identity, logo vectors, information flyers, and community outreach postcards.
- **Press Releases:** Published five major media updates:
    1. November 2021: *"Augmented Reality in Wind Energy Planning – Public Participation 2.0?"*
    2. April 2022: *"First Tier Field Evaluation Initiated: Launching a Novel Spatial App Prototype"*
    3. June 2022: *"Wind Infrastructure Visualized – ar4wind Commences Comprehensive Field Testing"*
    4. February 2023: *"Advanced Phase Field Trials Slated for February and March 2023"*
    5. April 2023: *"Phase II Public Evaluation Trials Successfully Concluded"*
- **Newsletters:** Distributed three regular progress updates to subscribers in July 2022, March 2023, and June 2023.
- **Trade Fair Presence:**
    - September 15–16, 2021: Live interactive app demonstrations at the FA Wind exhibition stand during the HusumWind 2021 exposition, facilitating deep discussions regarding public engagement barriers.
    - September 29, 2022: Delivered the presentation *"Mitigating Regulatory Objections via Real-Time Immersive Tracking"* at the German Wind Energy Association (BWE) booth during the WindEnergy Hamburg 2022 trade fair.
    - June 06, 2023: Live interactive software demonstrations at the LandPlan OS exhibition booth during the Central German Renewable Energy Industry Days.